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Course: Applied Physics

Lecture 04

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1. Motion in One Dimension (1D Motion)

Definition:

Motion in one dimension means an object moves along a straight line only. It has **only one coordinate** to describe its position (usually x-axis).

$$\vec{r} = x\hat{i}$$

Only one variable changes (x).

Examples:

- A train moving on a straight railway track
- A stone falling vertically under gravity
- A car moving on a straight road

2. Motion in Two Dimensions (2D Motion)

Definition:

Motion in two dimensions means an object moves in a plane. Its position is described using **two coordinates** (x and y).

$$\vec{r} = x\hat{i} + y\hat{j}$$

Both x and y change with time.

Examples:

- Projectile motion (a ball thrown in air)
- Circular motion in a plane
- A football kicked at an angle

3. Motion in Three Dimensions (3D Motion)

Definition:

Motion in three dimensions means an object moves in space. Its position requires **three coordinates** (x, y, z).

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

All three coordinates change with time.

Examples:

- A bird flying in the sky
- An airplane moving in air
- A satellite orbiting Earth

Comparison

Type of Motion	Coordinates Needed	Example
1D	x	Train on straight track
2D	x, y	Projectile motion
3D	x, y, z	Bird flying

4. Equations of Motion

| First Equation

$$v = u + at$$

Meaning: Final velocity = initial velocity + change due to acceleration.

| Second Equation

$$s = ut + \frac{1}{2}at^2$$

Meaning: Displacement = motion due to initial velocity + motion due to acceleration.

| Third Equation

$$v^2 = u^2 + 2as$$

Meaning: Connects velocity and displacement (time not included).

5. Position Vector ($\vec{r}$)

➤ The **position vector** of a particle is a vector that gives the position of the particle **with respect to the origin** of a coordinate system.

➤ It tells us **where the object is located in space**.

✓ In One Dimension (1D):

$$\vec{r} = x\hat{i}$$

Here:

- x = distance from origin along x-axis
- $\hat{i}$ = unit vector along x-axis

✓ In Two Dimensions (2D):

$$\vec{r} = x\hat{i} + y\hat{j}$$

Here:

- x = position along x-axis
- y = position along y-axis
- $\hat{i}, \hat{j}$ = unit vectors

✓ In Three Dimensions (3D):

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Here:

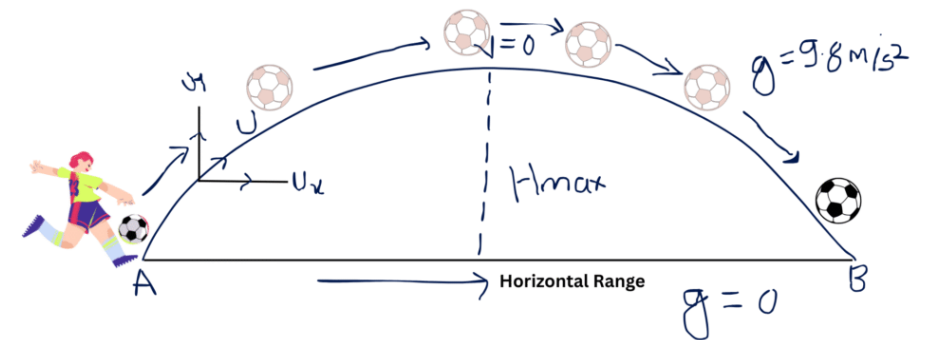
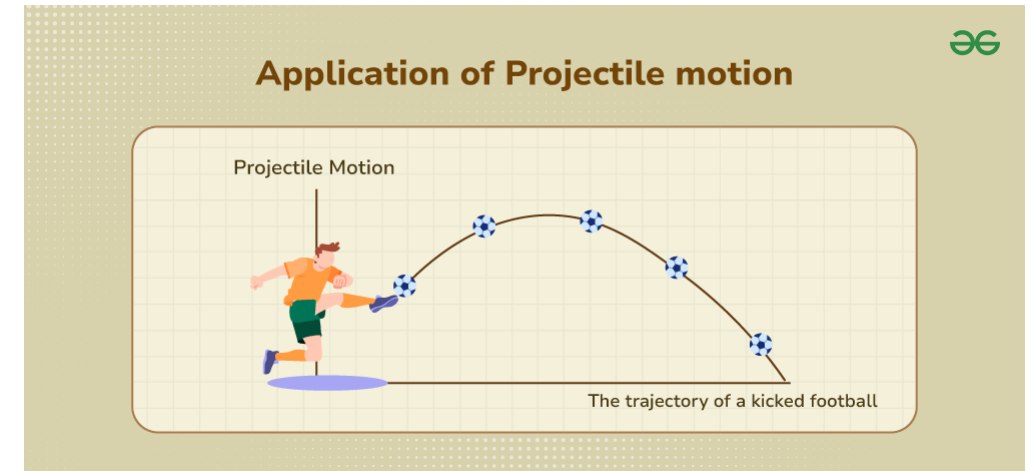
- x, y, z = coordinates of the particle
- $\hat{i}, \hat{j}, \hat{k}$ = unit vectors along x, y, z axes

6. Projectile Motion

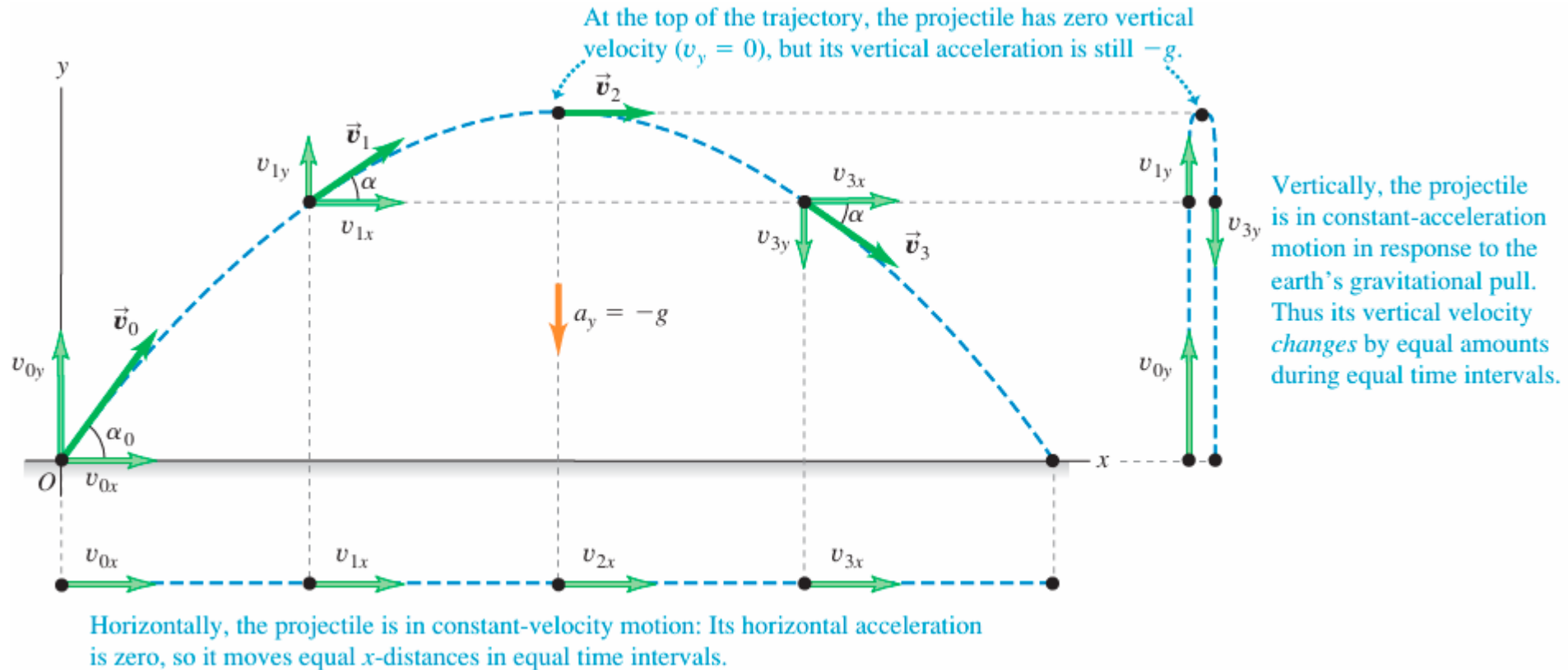
- Projectile motion is the motion of an object that is thrown or projected into the air and then moves only under the influence of gravity (air resistance neglected).
- Motion Under gravity.
- But generally it is considered as the motion in plane (plane is 2D surface)

Examples:

- A ball thrown at an angle
- A stone thrown horizontally from a building
- A bullet fired from a gun
- A football kicked into the air
- Once the object is projected, no force acts on it except gravity



Projectile Motion (2)



Projectile Motion (3)

x	y
$u_x = u \cos \theta$ <i>initial velocity in x direction</i>	$u_y = u \sin \theta$ <i>initial velocity in y direction</i>
$a_x = 0$ acceleration in x direction because velocity is constant	$a_y = -g$ (<i>-sign because it is downward</i>) acceleration in y direction In first, velocity decreases v_y and at Max. $v_y = 0$ and after that it increases

Time of Flight

The time for which an object remains in air.

Consider the motion in Y direction,

$$u_y = u \sin \theta$$

$$a_y = -g$$

$$u_y = u \sin \theta$$

$$v_y = 0$$

Now let's suppose equation of motion under uniform acceleration which is the First equation of motion

$$v = u + at$$

$$0 = u \sin \theta + (-g)t$$

$t = \frac{u \sin \theta}{g}$ so the total time of flight will be

$$t = \frac{2u \sin \theta}{g}$$

Maximum Height

- Maximum Height

Using the equation of motion:

$$s = ut - \frac{1}{2}gt^2$$

For vertical motion:

$$h_{max} = (u \sin \theta)t - \frac{1}{2}gt^2$$

At maximum height:

$$v = 0$$

Using:

$$v = u \sin \theta - gt$$

$$0 = u \sin \theta - gt$$

$$t = \frac{u \sin \theta}{g}$$

Maximum Height (2)

Substitute value of t in height equation

$$h_{max} = (u \sin \theta) \left(\frac{u \sin \theta}{g} \right) - \frac{1}{2} g \left(\frac{u \sin \theta}{g} \right)^2$$

Simplify

$$h_{max} = \frac{u^2 \sin^2 \theta}{g} - \frac{1}{2} \frac{u^2 \sin^2 \theta}{g}$$

$$h_{max} = \frac{u^2 \sin^2 \theta}{2g}$$

Final Result

$$h_{max} = \frac{u^2 \sin^2 \theta}{2g}$$

Equation of Trajectory

$$x = (u \cos \theta)t$$

$$t = \frac{x}{u \cos \theta}$$

Vertical Motion

$$y = (u \sin \theta)t - \frac{1}{2}gt^2$$

$$y = (u \sin \theta) \left(\frac{x}{u \cos \theta} \right) - \frac{1}{2}g \left(\frac{x}{u \cos \theta} \right)^2$$

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

Final Equation of Trajectory

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

This equation is of the form:

$$y = ax - bx^2$$

which represents a **parabola**.

So, the **path of a projectile is parabolic**.

The Horizontal Range

Horizontal motion:

$$x = (u \cos \theta)t$$

At total time of flight T , horizontal distance is the **range (R)**:

$$R = (u \cos \theta)T$$

Time of flight

From vertical motion:

$$T = \frac{2u \sin \theta}{g}$$

Substitute T in range equation

$$R = (u \cos \theta) \left(\frac{2u \sin \theta}{g} \right)$$

Simplify

$$R = \frac{2u^2 \sin \theta \cos \theta}{g}$$

Using identity:

$$2 \sin \theta \cos \theta = \sin 2\theta$$

Final Range Formula

$$R = \frac{u^2 \sin 2\theta}{g}$$

- Maximum range occurs when:

$$\theta = 45^\circ$$

- Maximum range value:

$$R_{max} = \frac{u^2}{g}$$